

Inaction Regions in Perpetual Futures: Deadband Funding and Price Convergence

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Abstract

Perpetual futures combine a contractually specified funding transfer with market-based price adjustment. I study a deadband architecture used by a Commodity Futures Trading Commission-regulated cryptocurrency perpetual-futures venue and compare its realised cash flows with thirteen corresponding Bybit contracts. In a common event-index window (Kalshi: 3 June 2026 20:00 UTC–11 July 2026 04:00 UTC; Bybit: four hours earlier), 73.0 per cent of 1,226 Kalshi settlements are exactly zero, compared with none of 1,469 offshore settlements. A three-day circular block bootstrap gives a 95 per cent interval of 65.7–80.4 per cent for Kalshi’s pooled zero share. Conditional on activation, however, the mean absolute Kalshi transfer is 1.88 basis points per eight hours, versus 0.73 basis points offshore. The deadband therefore creates an inaction region in contractual cash flows rather than merely lowering average funding. High-frequency evidence is consistent with premium mean reversion while funding remains inactive, with most observed adjustment borne by the perpetual mark rather than the reference index. The venue’s filed impact-price rule, user-facing candle description, and public mark-index proxy are not equivalent, although the proxy recovers the corrective sign and aggregate activation frequency. The results show how continuous market-based convergence can coexist with intermittent settlement transfers, and why derivative design should be evaluated through both activation frequency and conditional cash-flow magnitude.

Keywords: perpetual futures; funding rate; inaction region; derivative security design; market microstructure; price convergence

JEL classification: G13, G14, G23, D47

1 Introduction

A perpetual future has no maturity date and therefore no terminal settlement that mechanically forces its price to meet the underlying. Its design instead combines market-based adjustment

through trading, quoting, margin, and liquidation with a contractually specified funding cash flow between long and short positions. The frequency and state dependence of that cash flow determine not only average carry, but also when the contract intervenes and how funding risk is distributed through time.

Formal models treat funding as the cash-flow term that supports pricing and replication of a claim with no maturity (Angeris et al., 2022; He et al., 2022; Akerer et al., 2026). Empirical work has largely studied offshore venues on which funding is almost continuously nonzero, focusing on price discovery, arbitrage, market structure, carry, and high-frequency trading (Alexander et al., 2020; Aleti and Mizrach, 2021; De Blasis and Webb, 2022; Alexander et al., 2024; Schmeling et al., 2026; Shynkevich, 2026). This paper examines a materially different contract design. Kalshi, a designated contract market regulated by the Commodity Futures Trading Commission (CFTC), launched cryptocurrency perpetual futures in 2026 and suppresses the funding transfer when the rule input lies within a specified deadband (Commodity Futures Trading Commission, 2026a,b).

The deadband creates an inaction region in the contract’s settlement cash flow. It does not prevent trading, but it sets the transfer to zero until a premium-related state variable crosses a boundary. The design is analogous to inaction regions in transaction-cost models, where adjustment is delayed until the benefit exceeds a threshold (Davis and Norman, 1990). The economic object is nevertheless different: portfolio no-trade regions suppress investor rebalancing, whereas the mechanism studied here suppresses a contractual transfer while market participants remain free to trade. This distinction yields a direct derivative-design question: can continuous price convergence coexist with intermittent funding, and what cash-flow distribution does the threshold create?

The first result is a sharp regime contrast. During a matched event-index window, 895 of 1,226 Kalshi funding settlements are exactly zero, a share of 73.0 per cent. A three-day circular block bootstrap gives a 95 per cent interval of 65.7–80.4 per cent. None of 1,469 settlements for thirteen corresponding Bybit perpetuals is exactly zero. The contrast is not driven by uniformly smaller Kalshi rates. Conditional on activation, Kalshi’s event-weighted mean absolute rate is 1.88 basis points per eight-hour interval, compared with 0.73 basis points offshore. The threshold converts nearly continuous low-amplitude carry into sparse but comparatively lumpy transfers.

The second result concerns convergence inside the contractual inaction region. The public-endpoint premium mean reverts smoothly on both sides of zero, including over deviations associated with inactive funding. Conditional mark returns move strongly against the initial premium, whereas subsequent index returns are small. The observed short-horizon correction is therefore primarily mark-led. This evidence is descriptive rather than causal, but it is consistent with a two-layer mechanism: trading and quoting correct temporary deviations continuously, while funding responds only to deviations sufficiently persistent to survive the eight-hour input construction and deadband.

The third contribution concerns implementation and measurement. The filed contract terms specify a per-second Premium Index constructed from hypothetical impact bid and ask prices and combined with increasing chronological weights. The user-facing Help Center describes 480

one-minute candlestick premiums. The public mark-index proxy used here is a third representation. It recovers the corrective sign of all 23 activated complete-coverage settlements and reproduces aggregate activation within one percentage point, but it is not the filed input and its empirical slope is not a documented rate coefficient. The distinction matters because a mechanism audit must separate legal specification, user disclosure, observable fields, and realised cash flows.

The paper develops four testable implications. A deadband should create a larger point mass at zero than a continuous rule. Conditional transfers may be larger because small calculated rates are suppressed. Premium convergence may remain active inside the contractual inaction region. Finally, if the premium is a contract-specific dislocation rather than information about the future index, the mark should bear most of the subsequent adjustment. The data support all four implications descriptively, while the short sample and incomplete filed-rule reconstruction limit structural and causal claims.

The remainder of the paper proceeds as follows. Section 2 describes funding as state-contingent contract design and derives the testable implications. Section 3 presents the data and measurement. Section 4 compares realised funding cash flows. Section 5 examines activation and the public proxy. Section 6 studies convergence inside the contractual inaction region. Section 7 presents measurement diagnostics and limitations. Section 8 discusses implications for derivative security design, and Section 9 concludes.

2 Funding as State-Contingent Contract Design

2.1 Contractual and market-based convergence

Let M_t denote the contract mark and I_t the reference index. Define a public-endpoint premium in basis points as

$$p_t = 10^4 \frac{M_t - I_t}{I_t}. \quad (1)$$

A positive premium means the contract is rich relative to the index. A conventional funding rule maps a premium-related input into a transfer r_T at settlement time T . A positive r_T is paid by longs to shorts; a negative r_T is paid by shorts to longs. The intended corrective sign is therefore

$$\text{sign}(r_T) = \text{sign}(\text{funding input}_T). \quad (2)$$

Let x_T denote the mechanism's premium-related input, $\tau > 0$ its deadband, and $g(\cdot)$ the transfer mapping outside the band. A generic thresholded rule is

$$r_T = \begin{cases} 0, & |x_T| < \tau, \\ g(x_T), & |x_T| \geq \tau. \end{cases} \quad (3)$$

Equation (3) creates an inaction region in settlement cash flows. It does not create a no-trade region: the mark, index, and positions can continue to adjust while $r_T = 0$.

Funding is not the only force on the premium. Traders may buy the cheap leg and sell the expensive leg, liquidity suppliers may revise quotes, and liquidation or margin constraints can alter inventory. In reduced form, short-horizon premium dynamics may be written as

$$E[\Delta p_{t,t+h} \mid p_t] = \mu(p_t), \quad (4)$$

where convergence implies $\mu(p) > 0$ for $p < 0$ and $\mu(p) < 0$ for $p > 0$. If market-based adjustment operates continuously, equation (4) can hold even when the contractual funding rate is zero.

2.2 Kalshi’s thresholded design

Kalshi’s filed BTCPERP terms specify a Premium Index based on the underlying index and the average execution prices of hypothetical \$1,000 impact orders on the bid and ask sides (KalshiEX LLC, 2026). The \$1,000 Impact Notional is prescribed by the exchange—and described in the filing as an approximation to median trade size—rather than selected by the researcher. At each sampled second, the impact prices depend on contemporaneous live order-book depth; a side is undefined when an order of the prescribed size cannot be filled in full. The filed Premium Index is sampled once per second over an ordinarily eight-hour period, with a uniformly selected instant within each one-second range. Later observations receive linearly increasing weights, and the filing expressly distinguishes this construction from an equal-weighted TWAP. The filed deadband sets the rate to zero when the absolute Weighted Average Premium Index (WAPI) is below 0.01 per cent (one basis point), and a separate rule sets an undeterminable rate to zero, including where order-book liquidity is insufficient.

The venue’s Help Center describes the input differently: a time-weighted average of 480 one-minute candlestick premiums, followed by a 0.01 per cent zero threshold in the calculated funding rate and a cap (Kalshi, 2026). The two descriptions share the eight-hour horizon and a one-basis-point official threshold, but differ in their observable input and averaging construction. The public settlement-mark/index premium in equation (1) is a third representation and is measured in premium basis points, not funding-rate basis points.

2.3 Offshore comparison

Bybit’s documented rule uses a linearly weighted 480-minute Premium Index together with an interest-rate component and clamps (Bybit, 2026b). In the matched sample, the contracts settle every eight hours. Bybit does not impose Kalshi’s exact-zero deadband, so small nonzero transfers are common.

Table 1 summarises the relevant differences. The comparison isolates a realised regime contrast,

not a causal regulatory treatment. The venues differ in age, liquidity, participant access, index construction, mark formation, and funding formula.

Table 1: Funding-design representations used in the paper

Feature	Kalshi filed terms	Kalshi Help Center	Bybit documentation
Core input	Reference index and hypothetical \$1,000 impact bid/ask prices; notional fixed by the exchange	One-minute candlestick premiums	Weighted Premium Index and interest component
Averaging	Per-second observations with increasing chronological weights	480 one-minute observations over eight hours	Linearly weighted 480-minute Premium Index
Exact-zero rule	$ WAPI < 0.01\%$ (one basis point); undeterminable rate also set to zero	Absolute calculated rate below 0.01% (one basis point) set to zero	No corresponding exact-zero deadband
Settlement cadence	Eight hours	Eight hours	Eight hours for matched contracts
Empirical object used here	Realised settlements and a public settlement-mark/index proxy; neither documented input is directly observed in the exploratory sample		Public settled funding history

Threshold units used throughout the paper. The official filed deadband is $|WAPI| < 1$ basis point, and the Help Center describes a one-basis-point threshold in the calculated funding rate. The approximately ± 3 basis-point region reported below is an empirical mapping into the eight-hour mean *public mark-index premium*; it is not an official rule parameter. Bands at ± 2 or ± 2.5 basis points in exploratory plots are descriptive visual aids only. Before the deadband and clamp, the filed WAPI is itself the funding rate; the slope relating the public proxy to realised funding is therefore an empirical cross-representation mapping, not a documented rate coefficient.

2.4 Testable implications

The contract design yields four testable implications.

Implication 1: a point mass at zero. The exact-zero share should be materially larger under the deadband than on an offshore venue without the same rule.

Implication 2: larger conditional transfers. Suppressing small rates can lower unconditional funding frequency while increasing $E(|r_T| \mid r_T \neq 0)$ relative to a continuous rule.

Implication 3: convergence within the contractual inaction region. If market-based trading and quoting correct temporary deviations continuously, equation (4) should show mean reversion even for premiums inside the region associated with zero funding.

Implication 4: mark-led correction. If the premium reflects a dislocation specific to the perpetual contract rather than information in the contract about the future index, the mark should move toward the index more strongly than the index moves toward the mark.

3 Data and Measurement

3.1 Kalshi data

The exploratory collector queried public production endpoints without authentication. It recorded the all-contract market-state snapshot at five-second cadence, per-contract order books at two-second cadence, the published funding-rate estimate at sixty-second cadence, and settled funding history hourly. Raw responses were retained alongside parsed fields. The order-book snapshots are live observations to which the exchange-defined \$1,000 Impact Notional can be applied; they are not fixed-depth assumptions. Because they arrive every two seconds rather than at the filing’s randomly selected instant within every second, they support only an approximate retrospective reconstruction of the filed input.

The high-frequency market-state window runs from 11 July 2026 06:03 UTC to 25 July 2026 07:07 UTC. The snapshot endpoint returned all sixteen contracts together, producing 2,238,768 observations, exactly 139,923 per contract. At nominal five-second cadence these observations correspond to approximately 194.3 hours, or 57.7 per cent of the 337.1-hour wall-clock window. The lost coverage arose from collector-host reboots, a stopped process, and host crashes rather than market events.

Settled funding history is not limited by these interruptions because the history endpoint returns past settlements. The full settlement sample contains 1,759 observations across thirteen contracts from 3 June through 24 July 2026. Three listed contracts have no settled-funding history and are discussed in Appendix B.

3.2 Contract-interval sample

The proxy analysis uses one observation per contract-settlement interval. For settlement time T , define the eight-hour mean public premium

$$\bar{p}_T = \frac{1}{N_T} \sum_{t \in [T-8h, T)} p_t. \quad (5)$$

An interval is retained when each of its eight constituent hours contains at least 300 usable premium observations. Of 364 intervals with any usable data, 260 satisfy this complete-coverage rule. A fully observed interval contains approximately 5,750 observations; the median retained interval contains 5,751.

The sixty-second funding-estimate polls are not treated as independent observations. Each

contract-interval contributes hundreds of nearly identical estimates of one eventual settlement. Pooling those polls would create pseudo-replication and severely understate uncertainty.

3.3 Matched offshore sample

For the external benchmark, settled funding histories are collected from Bybit’s public V5 endpoint for thirteen USDT perpetuals corresponding to the Kalshi contracts with settlement histories (Bybit, 2026a). The Kalshi benchmark sequence runs from 3 June 2026 20:00 UTC through 11 July 2026 04:00 UTC. The corresponding Bybit sequence runs from 3 June 2026 16:00 UTC through 11 July 2026 00:00 UTC. Kalshi settles at 04:00, 12:00, and 20:00 UTC, while the matched Bybit contracts settle at 00:00, 08:00, and 16:00 UTC. The analysis preserves this four-hour event-index offset and compares distributions; it does not treat the observations as synchronous responses to identical market states.

An exact zero is numerical equality to zero in the public history response after parsing; no tolerance band is imposed. Rates are placed on a common basis-point-per-eight-hour scale.

3.4 Dependence-robust uncertainty

Funding outcomes are serially dependent within contracts and contemporaneously dependent across contracts. For the pooled Kalshi zero share, uncertainty is estimated using a circular block bootstrap with three-day blocks. Each resampled block retains the contemporaneous cross-section of contracts, preserving common market shocks as well as within-contract clustering. Contract-level intervals are reported in Appendix E.

4 Realised Funding Cash Flows

4.1 Full Kalshi settlement history

Across the full settlement history, 1,391 of 1,759 observations are exactly zero, or 79.1 per cent. Negative rates account for 15.9 per cent and positive rates for 5.1 per cent. Negative funding is paid by shorts to longs, so nonzero transfers favour long positions by roughly three to one. Seven of thirteen contracts never charge longs during their available histories. Bitcoin and Chainlink have positive cumulative funding, but Chainlink’s cumulative rate is economically negligible relative to Bitcoin’s.

The asymmetry is descriptive. It may reflect directional demand, venue-specific capital constraints, differences between the participant base and the offshore exchanges contributing to the reference index, or limits to arbitrage (Duffie, 2010; Gromb and Vayanos, 2010; Makarov and Schoar, 2020). The present data do not identify the mechanism.

Table 2: Settled Kalshi funding by contract, 3 June–24 July 2026

Contract	n	Ann. mean (%)	Cumulative	Positive (%)	Zero (%)	Negative (%)	Longest non-positive run
KXBCHPERP	136	−7.82	−0.00971	0.0	73.5	26.5	136
KXBTCPERP	154	+4.13	+0.00581	27.9	71.4	0.6	30
KXDOGEPPERP	136	−1.13	−0.00140	0.0	93.4	6.6	136
KXETHPERP	154	−2.53	−0.00356	2.6	81.8	15.6	82
KXHYPEPPERP	139	−0.38	−0.00048	7.2	71.9	20.9	77
KXKSHIBPERP	136	−6.10	−0.00757	0.0	68.4	31.6	136
KXLINKPERP	138	+0.17	+0.00021	2.2	97.1	0.7	134
KXLTCPERP	136	−3.56	−0.00442	0.0	75.7	24.3	136
KXNEARPERP	93	−2.46	−0.00209	0.0	84.9	15.1	93
KXSOLPERP	154	−1.34	−0.00188	0.6	94.8	4.5	81
KXSUIPERP	136	−4.56	−0.00566	0.0	71.3	28.7	136
KXXRPPERP	154	−1.45	−0.00204	18.2	68.2	13.6	76
KXZECPERP	93	−4.25	−0.00361	0.0	76.3	23.7	93
All	1,759			5.1	79.1	15.9	

Notes: The annualised mean is the simple rate $1,095\bar{r}$, not a realised sample return or compounded annual rate. Positive funding is paid by longs to shorts. “Longest non-positive run” is the longest sequence over which longs were never charged.

4.2 Gross contractual carry versus trading profitability

For unit notional, the cumulative funding cash flow to a continuously held long is

$$C_{i,[0,T]}^L = - \sum_{t=1}^T r_{i,t}, \quad (6)$$

under the sign convention used here. The cumulative column in Table 2 reports $\sum r_{i,t}$, so a negative entry is a gross receipt to the long side. This is contractual carry, not an executable arbitrage return. A cash-and-carry strategy would additionally require bid–ask spreads, trading fees, spot financing or borrow costs, margin, settlement timing, and liquidation risk. The paper therefore quantifies the timing and sign of funding cash flows without claiming net arbitrage profitability.

4.3 Matched Kalshi–Bybit comparison

Implication 1 receives direct support. In the matched event-index window, Kalshi records 895 exact zeros in 1,226 prints, a zero share of 73.0 per cent. Kalshi timestamps run from 3 June 2026 20:00 UTC through 11 July 2026 04:00 UTC; the Bybit benchmark timestamps run four hours earlier. The three-day circular block-bootstrap interval is 65.7–80.4 per cent. None of 1,469 Bybit prints across the corresponding contracts is exactly zero.

Figure 1 shows the point mass at zero. The shaded band marks Kalshi’s documented one-basis-point threshold in settled-rate units. Figure 2 illustrates the temporal difference for Bitcoin: Bybit changes by small amounts almost every interval, while Kalshi alternates between zero runs and occasional larger transfers. The series are not synchronous pairs because their settlement times differ by four hours.

Implication 2 is also supported. Conditional on a nonzero print, the event-weighted mean absolute Kalshi rate is 1.88 basis points, compared with 0.73 basis points on Bybit. The relationship holds

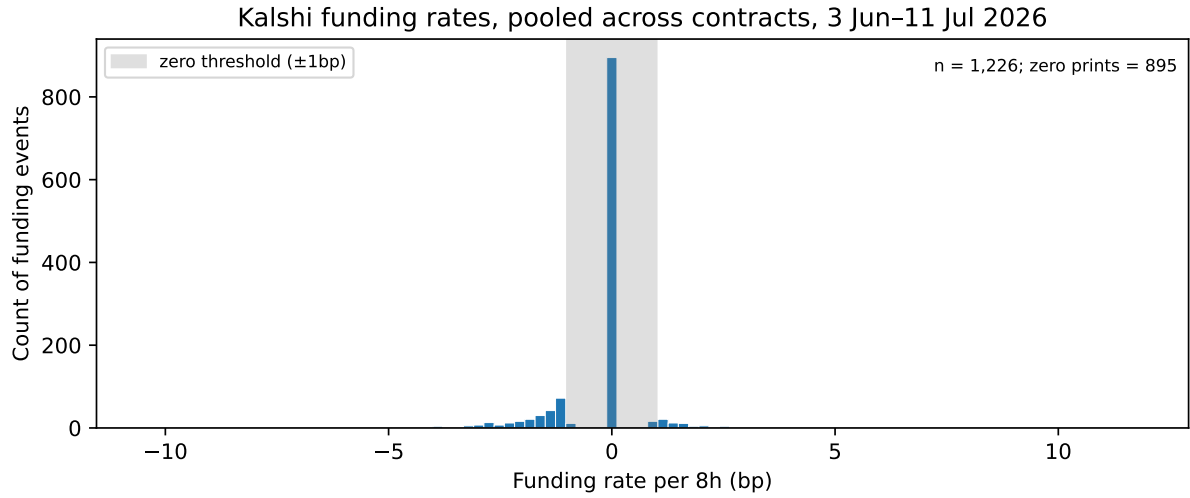


Figure 1: Distribution of Kalshi funding rates in the matched window (Kalshi settlement sequence: 3 June 2026 20:00 UTC–11 July 2026 04:00 UTC). The sample contains 1,226 settlements across thirteen contracts, including 895 exact zeros. The shaded region marks the official one-basis-point threshold in the calculated funding-rate/WAPI representation; it is not a band on the public mark-index premium.

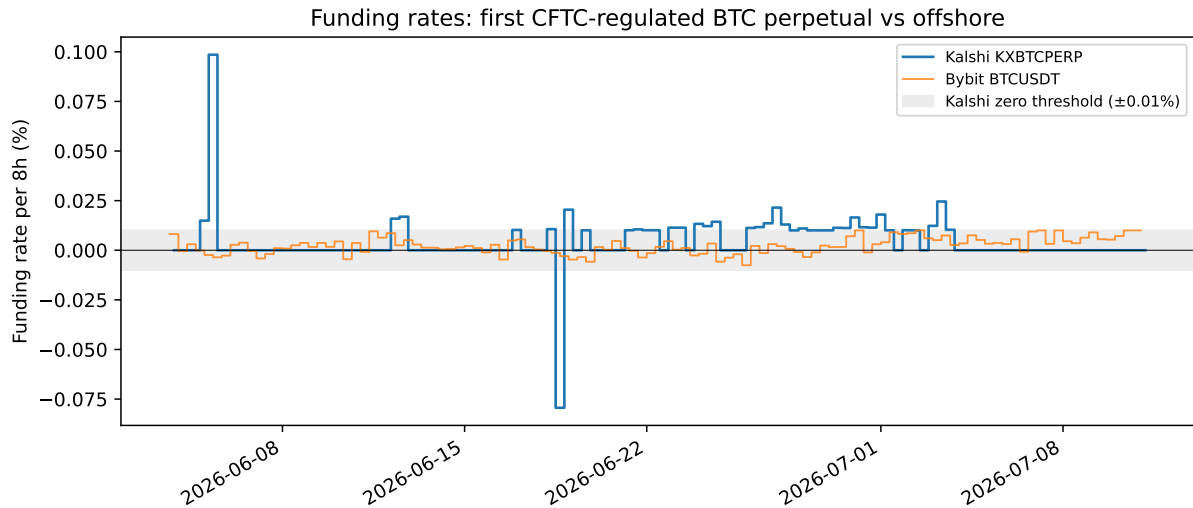


Figure 2: Bitcoin funding rates on Kalshi and Bybit. The Kalshi sequence runs from 3 June 2026 20:00 UTC through 11 July 2026 04:00 UTC; the Bybit sequence runs four hours earlier. Both are expressed per eight-hour interval. The shaded region is Kalshi’s official one-basis-point funding-rate/WAPI threshold and is shown only as a reference for Bybit.

for twelve of the thirteen matched assets; Zcash is the exception. Thus the deadband does not simply compress funding. It removes small transfers and leaves a distribution with less frequent but larger realised movements.

Table 3: Matched event-index funding outcomes (Kalshi: 3 June 2026 20:00 UTC–11 July 2026 04:00 UTC; Bybit: four hours earlier)

Asset	Kalshi			Bybit		
	n	Zero (%)	Mean $ r $ if nonzero (bp)	n	Zero (%)	Mean $ r $ (bp)
BCH	95	62.1	2.70	113	0.0	0.79
BTC	113	64.6	1.65	113	0.0	0.38
DOGE	95	90.5	1.56	113	0.0	0.53
ETH	113	77.0	1.84	113	0.0	0.39
HYPE	98	64.3	2.06	113	0.0	0.60
SHIB	95	54.7	1.76	113	0.0	0.62
LINK	97	99.0	1.45	113	0.0	0.56
LTC	95	67.4	1.35	113	0.0	0.57
NEAR	52	80.8	1.42	113	0.0	0.73
SOL	113	92.9	2.63	113	0.0	0.74
SUI	95	65.3	1.51	113	0.0	0.68
XRP	113	56.6	2.14	113	0.0	0.52
ZEC	52	80.8	1.67	113	0.0	2.34
Pooled	1,226	73.0	1.88	1,469	0.0	0.73

Notes: Rates are basis points per eight-hour interval. The Kalshi conditional mean excludes exact zeros. The Bybit mean is unconditional because no matched print is exactly zero. Kalshi contracts enter when their settlement histories begin, so n varies by asset.

The comparison is not a causal estimate of regulation. The result establishes a venue-specific funding regime in this sample. It does not determine whether the difference is caused by the deadband alone, by other formula components, or by venue liquidity and participation.

5 Funding Activation and the Observable Proxy

5.1 Proxy association and corrective sign

The complete-coverage sample pairs each settled rate with the preceding eight-hour mean public mark-index premium. Across the 23 activated intervals, regressing the settled rate on $\bar{p}_T$ gives a slope of 3.24×10^{-5} per basis point and a correlation of 0.967. Equivalently, a one-basis-point change in the public proxy is associated with approximately 0.324 basis points of settled funding within this selected activated sample. All 23 activated rates have the corrective sign predicted by the proxy.

The filed terms do not specify this empirical slope. Before the deadband and clamp, the filed WAPI is itself the funding rate. The smaller public-proxy relationship therefore cannot validate documented rate scaling; it reflects the use of a different price input, averaging construction, and sampling process, together with possible attenuation from endpoint alignment. The statistics show that the public proxy contains relevant economic signal, not that it reconstructs the production formula.

5.2 Activation function

As an exploratory cross-representation mapping, dividing the official one-basis-point WAPI/funding-rate threshold by the estimated 0.324 coefficient implies an eight-hour mean public-premium boundary of approximately 3.1 basis points. The coefficient is estimated from only 23 activated intervals and is not structural. Applying a three-basis-point public-proxy threshold to the 260 complete-coverage intervals predicts activation in 9.8 per cent, compared with 8.8 per cent realised. This aggregate agreement does not establish interval-level implementation fidelity.

Figure 3 displays the activation function. Activation is nearly absent around zero and increases on both sides outside the central region. The curve is economically consistent with a deadband. Its imperfect determinism is also informative: the public proxy is not the production input.

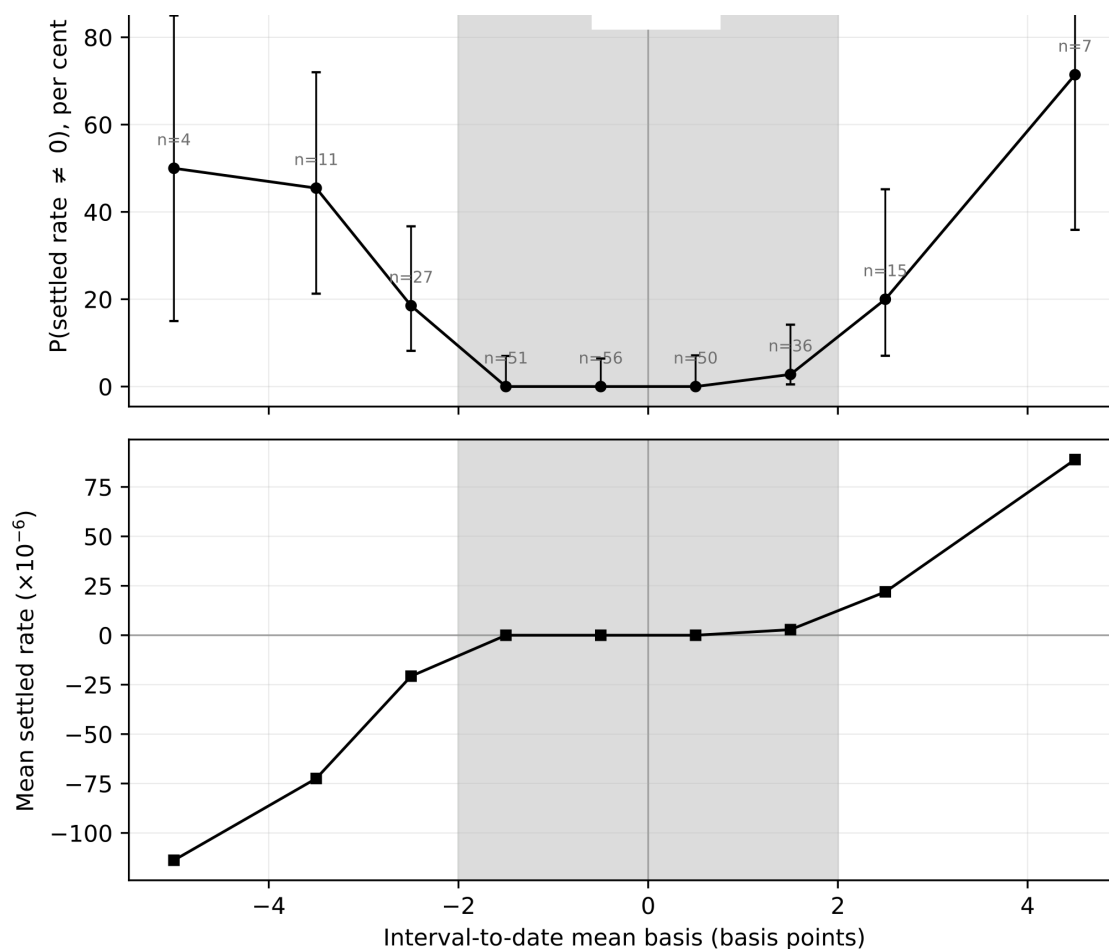


Figure 3: Funding activation and magnitude against the eight-hour mean public-endpoint premium. The plotted bins cover 257 of 260 complete-coverage contract-intervals and 21 of 23 activations; three tail observations, including two activations, lie outside the displayed range. Bars are Wilson 95 per cent intervals. The shaded ± 2 basis-point region is descriptive only. Mapping the official one-basis-point calculated-rate threshold through the empirical proxy slope gives an approximate ± 3 basis-point public-premium region; neither value replaces the official rule parameter.

A direct implementation audit requires interval-level comparison of three candidate inputs: the filed impact-price construction, the Help Center candle construction, and the public proxy. The relevant outcomes are activation accuracy, false positives and false negatives, sign accuracy, and mean absolute rate error. The exploratory order-book archive may support an approximate filed-rule reconstruction, but it cannot reproduce every random within-second sample described in the filing.

6 Price Convergence Inside the Contractual Inaction Region

6.1 Nonparametric premium drift

Prediction 3 asks whether convergence continues inside the region in which funding is ordinarily zero. For each initial premium bin, I calculate the subsequent five-minute change in the premium and express it as an hourly drift. Figure 4 shows a smooth restoring pattern. Negative premiums are followed by positive premium changes, while positive premiums are followed by negative changes. The curve crosses zero close to a zero initial premium and displays no visible discontinuity at the illustrative deadband edge.

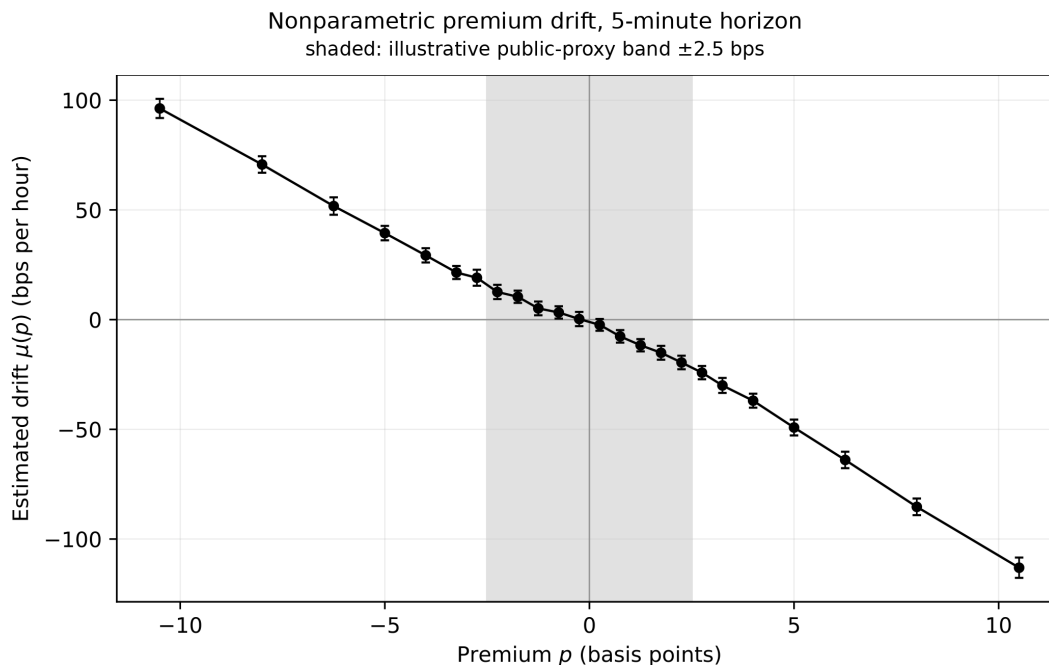


Figure 4: Nonparametric five-minute drift of the public-endpoint premium. Points are conditional bin means and bars are uncertainty intervals produced by the source analysis. The shaded ± 2.5 basis-point region is an illustrative public-proxy band, not the official one-basis-point WAPI/funding-rate threshold. The continuous restoring pattern supports short-horizon mean reversion inside as well as outside the region, but the figure is exploratory and does not identify the causal effect of funding.

The finding changes the interpretation of zero funding. A zero contractual transfer does not mean that the contract and index are disconnected. Temporary deviations can be corrected by trading

before they persist long enough to affect the eight-hour funding input. The pattern is consistent with funding operating as an intermittent contractual backstop layered on top of continuous market-based convergence.

6.2 Which price adjusts?

Implication 4 separates two possible channels. The premium can close because the mark moves toward the index, because the index later moves toward the mark, or because both respond. Figure 5 plots conditional fifteen-minute returns. The mark return is strongly decreasing in the initial premium: it is positive after a discount and negative after a premium. Conditional index returns are small and remain modestly negative across bins.

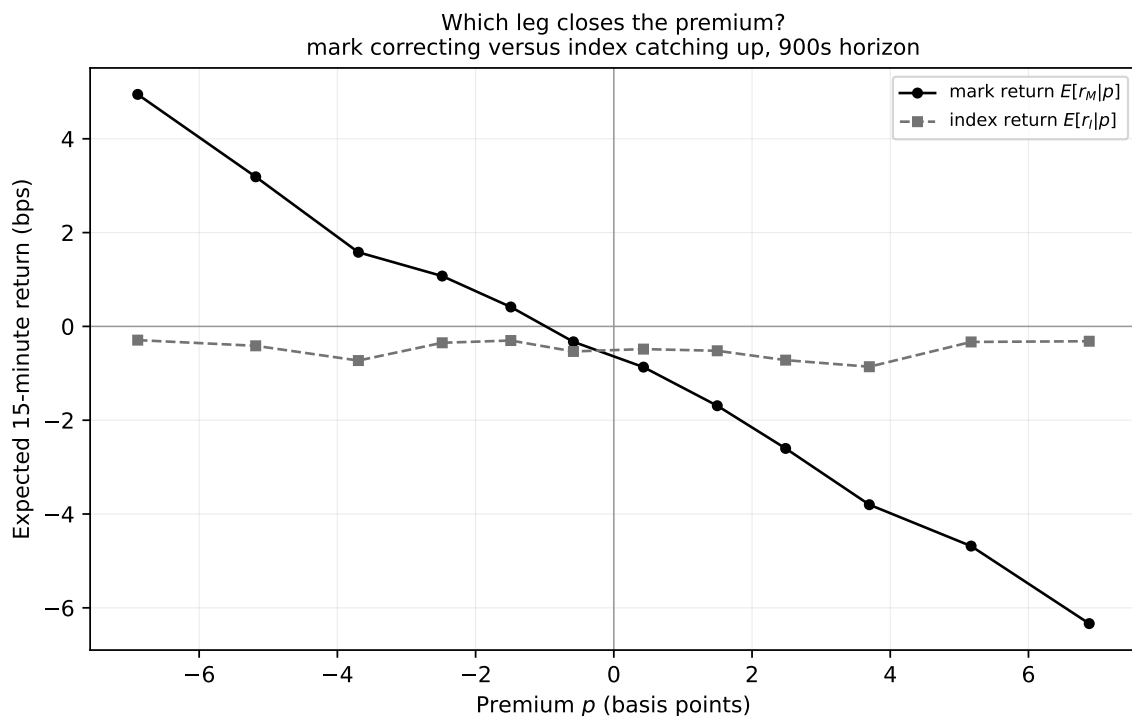


Figure 5: Conditional fifteen-minute mark and index returns as functions of the initial public-endpoint premium. The mark moves strongly against the initial deviation, while the index response is small. The calculation is descriptive and does not condition on eventual funding activation.

The observed short-horizon correction is predominantly mark-led. The result is consistent with the premium representing a contract-specific deviation that attracts liquidity or arbitrage, rather than the mark systematically forecasting subsequent index movements.

7 Measurement, Robustness, and Limitations

7.1 Endpoint alignment

The public market-state response timestamps the mark and index separately. In the high-frequency sample, the mark is a median 2.5 seconds stale at collection and the index 0.4 seconds stale; the standard deviation of their timestamp difference is 183 seconds. A signal-plus-noise decomposition estimates transitory premium dispersion between 2.6 and 6.6 basis points across contracts. This transitory component is strongly associated with five-second index-return volatility ($R^2 = 0.929$) and unrelated to tick size ($R^2 = 0.000$).

The pattern supports an asynchronous-sampling interpretation. It does not by itself identify asynchronicity structurally because the decomposition assumes limited serial dependence in the transitory component and an exponential persistent component. Appendix C reports the full diagnostics. The practical implication is narrower and robust: an instantaneous mark-index difference from public fields contains several basis points of alignment variation, while an eight-hour average removes much of it.

7.2 Alternative samples and inference

The point mass at zero survives dependence-robust inference. The pooled block-bootstrap interval of 65.7–80.4 per cent remains far from Bybit’s zero share. Contract-level intervals are wide for recently listed contracts, but every point estimate exceeds 54 per cent.

The activation curve also remains qualitatively similar when all 364 contract-intervals with any usable coverage are included. The preferred sample retains only 260 complete intervals because partial coverage can bias an eight-hour mean. Appendices D–G report both curves, the averaging-horizon diagnostics, and alternative premium-density displays.

7.3 Limitations

Six limitations bound the conclusions. First, the venue is young and the longest settlement history is 51 days. Second, the high-frequency collector covers 57.7 per cent of the nominal wall-clock sample. Third, the public mark-index premium is not the filed impact-price Premium Index. Fourth, the convergence analysis is exploratory and is not a quasi-experimental estimate of funding’s causal effect. Fifth, the offshore benchmark uses one venue and is calendar-matched rather than timestamp-synchronous. Sixth, the venues differ in liquidity, participants, index design, mark construction, and numerical reporting.

These limitations matter most for normative claims. The paper does not show that a deadband is superior, that regulation caused the observed regime, or that the filed implementation is exact. It shows that a thresholded venue produces a distinct realised funding distribution and that price convergence continues when the contractual transfer is inactive.

8 Implications for Derivative Security Design

The evidence supports a two-layer view of perpetual-futures convergence. At high frequency, the observed mark moves toward the reference index in a pattern consistent with adjustment through trading and quoting. At the settlement horizon, the funding contract transfers cash only when the rule input is sufficiently persistent and large. A long averaging window filters short-lived variation, and the deadband suppresses small calculated rates. The resulting security combines continuous tradability with an intermittent state-contingent cash flow.

This distinction matters for the design and valuation of perpetual contracts. Funding rules with similar unconditional means can expose positions to very different timing risk. Under continuous funding, carry is distributed across nearly every interval. Under a deadband, the same broad corrective objective may be implemented through long zero runs and concentrated payments. Average funding therefore does not describe the contract fully; the zero share, conditional magnitude, run length, tail concentration, and covariance with market stress are additional state variables relevant to risk management and hedging.

The design may also change risk sharing between directional traders and liquidity providers. Suppressing small transfers reduces routine carry paid by positions on the expensive side, but leaves larger cash flows when persistent deviations cross the threshold. Whether this improves welfare depends on transaction costs, capital constraints, liquidity, and the informational content of the input. The present evidence does not rank the mechanisms. It identifies a distinct cash-flow architecture and shows that contractual inactivity need not coincide with price-level inactivity.

The asymmetry of realised transfers is economically notable. Negative rates outnumber positive rates by about three to one, so longs are net recipients on most contracts. The pattern may reflect participant segmentation, constrained arbitrage, index construction, or directional demand (Duffie, 2010; Gromb and Vayanos, 2010; Makarov and Schoar, 2020). A longer panel with matched cross-venue basis, depth, and participant-side measures would be needed to determine whether the asymmetry is structural or specific to the launch period.

Finally, the disclosure comparison illustrates a general problem in empirical derivatives research. The filed rule, user-facing explanation, public data representation, and realised cash flow may occupy different layers of the mechanism. A public proxy can recover aggregate behaviour while misclassifying individual intervals. Conversely, a high-level disclosure can be economically accurate while omitting implementation details that matter near a threshold. Evaluation of a new derivative therefore requires both contract-level documentation and transaction- or order-book-level reconstruction.

9 Conclusion

This paper studies an inaction region in the funding cash flow of perpetual futures. In a common event-index window, 73.0 per cent of Kalshi settlements are exactly zero, compared with none for

thirteen corresponding Bybit contracts. Conditional on activation, Kalshi’s mean absolute rate is more than twice the offshore mean. The deadband therefore changes the temporal distribution of funding from continuous low-amplitude carry to sparse but comparatively lumpy transfers.

Contractual inactivity does not imply price-level inactivity. The public premium mean reverts within the region associated with zero funding, and the perpetual mark bears most of the observed short-horizon adjustment. The evidence is consistent with continuous market-based convergence operating alongside intermittent contractual settlement.

The public mark-index proxy recovers the corrective sign and aggregate activation frequency, but it is not equivalent to the filed impact-price rule or the Help Center candle description. A definitive implementation test requires interval-level reconstruction of all three representations in the preregistered holdout sample.

The broader implication is that funding frequency alone is not a sufficient measure of mechanism strength or contract risk. Perpetual-futures design should be evaluated through the joint distribution of zero frequency, conditional transfer size, run length, convergence speed, and the information used to calculate settlement. Deadbands create state-contingent cash-flow inaction, not necessarily market inaction.

Working-paper status

This manuscript is circulated as a working paper for scholarly discussion.

Statements and Declarations

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Ethics approval and consent. Not applicable. The study uses public market and exchange data and involves no human participants or animals.

Author contribution. The sole author conceived the study, designed and implemented the data collection, specified and conducted the analysis, verified the results, and wrote the manuscript.

Data, materials, and code availability. The datasets generated and analysed in this study consist of publicly sourced market, order-book, index, and funding data collected from Kalshi and Bybit. The processed datasets, collection code, database schema, and analysis scripts are available from the author upon reasonable request. Raw API responses were retained to permit verification of the reported variables and results, subject to repository file-size limits and venue terms.

Use of artificial intelligence. A large language model was used as a coding and editing assistant to help draft and debug collection and analysis code, convert the manuscript to L^AT_EX, and improve

prose clarity. The author designed the study, specified the analysis, ran and checked the code, verified the reported numbers, and accepts full responsibility for the manuscript.

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Appendices

A Collection Coverage

The high-frequency collector ran from 11 July 2026 06:03 UTC to 25 July 2026 07:07 UTC. Table 4 records the known host events. The durations sum to approximately 136.1 hours. Snapshot-response coverage calculated from unique nominal five-second observations is 57.7 per cent of the wall-clock interval. The difference between host-event time and missing nominal samples reflects request latency, process start and stop boundaries, and smaller unlogged gaps.

Table 4: Collection interruptions, 11–25 July 2026

From (UTC)	To (UTC)	Hours	Cause
16 Jul 17:07	18 Jul 01:20	32.2	Unattended operating-system reboot
18 Jul 02:26	19 Jul 23:20	44.9	Process stopped, not restarted
19 Jul 23:53	21 Jul 23:45	47.9	Unattended operating-system reboot
22 Jul 05:18	22 Jul 11:31	6.2	Host crash
23 Jul 01:45	23 Jul 05:54	4.2	Host crash
23 Jul 07:06	23 Jul 07:24	0.3	Transient
23 Jul 07:42	23 Jul 07:58	0.3	Transient
Total		136.1	

B Contracts without Funding Histories

Three of sixteen listed contracts—KXDOTPERP, KXHBARPERP, and KXXLMPERP—lack a published settlement-mark field in 88,565 of 139,923 market-state observations each, or 63.3 per cent. The identical count is consistent with a shared upstream or reporting condition. The public funding-history endpoint returns no records for these contracts, and each public funding estimate is exactly zero in all 11,721 collected observations.

These observations do not establish that the contractual mechanism was absent. The filed rule is based on impact bid and ask prices rather than the public settlement-mark field and sets an undeterminable rate to zero, including where depth is insufficient (KalshiEX LLC, 2026). The contracts are excluded from settlement statistics because no outcome is available.

C Signal–Noise Diagnostics

Write the observed public premium as $p_t = f_t + u_t$, with a persistent component f_t and transitory component u_t . If the transitory component has little serial dependence beyond short lags, the premium autocorrelation at longer lags can be written as $\rho_p(k) = A\rho_f(k)$, where A is the persistent variance share. An exponential is fitted to the premium autocorrelation over lags from 30 seconds

to ten minutes and extrapolated to lag zero. The implied dispersions are $\sigma_f = \sigma_p\sqrt{A}$ and $\sigma_u = \sigma_p\sqrt{1-A}$.

Table 5: Signal–noise decomposition of the public-endpoint premium

Contract	σ_p	A	σ_f	σ_u	Index vol.	Tick	σ_u/idx
KXBCHPERP	5.29	0.208	2.41	4.71	2.367	0.408	1.99
KXBTCPERP	8.81	0.916	8.43	2.55	1.465	0.156	1.74
KXDOGERP	4.34	0.331	2.49	3.55	1.789	0.135	1.98
KXETHPERP	17.99	0.970	17.72	3.14	1.871	0.557	1.68
KXHYPEPERP	6.38	0.528	4.64	4.39	2.504	0.151	1.75
KXKSHIBPERP	5.03	0.279	2.66	4.27	2.023	0.228	2.11
KXLINKPERP	4.79	0.386	2.98	3.76	1.935	0.125	1.94
KXLTCPERP	3.62	0.295	1.97	3.04	1.628	0.223	1.87
KXNEARPERP	7.52	0.399	4.75	5.83	3.066	0.525	1.90
KXSOLPERP	4.06	0.174	1.69	3.69	1.821	0.129	2.02
KXSUIPERP	5.94	0.556	4.43	3.96	1.992	0.134	1.99
KXXRPPERP	3.69	0.193	1.62	3.32	1.823	0.905	1.82
KXZECPERP	9.17	0.475	6.32	6.64	3.071	0.200	2.16

Notes: All dispersions are basis points. Regressing σ_u on five-second index-return volatility gives $R^2 = 0.929$ and Spearman +0.945; regressing it on tick size gives $R^2 = 0.000$ and Spearman 0.000.

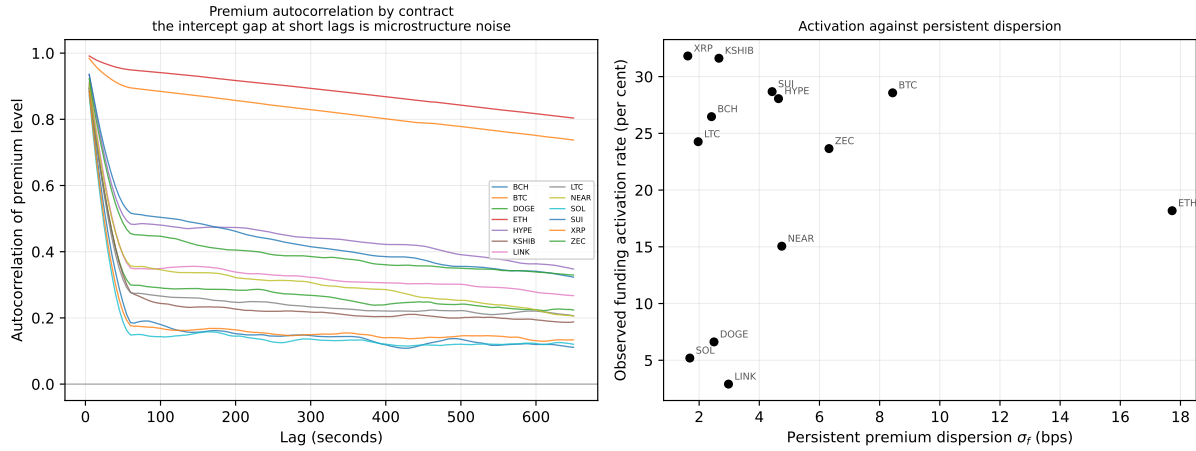


Figure 6: Premium autocorrelation by contract and contract-level activation against persistent premium dispersion. The right panel contains thirteen observations and is exploratory.

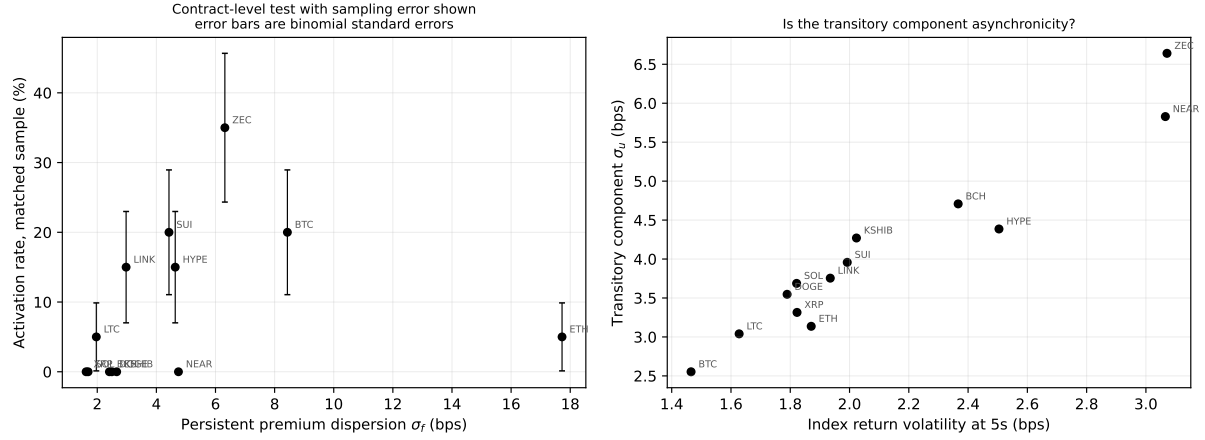


Figure 7: Contract-level activation with binomial standard errors and transitory premium dispersion against index-return volatility.

D Averaging Horizon

Figure 8 shows that the standard deviation of window means falls as the averaging horizon increases. The observed decline is slower than an independent-observation benchmark because the premium is persistent. The right panel applies illustrative one-, two-, and three-basis-point thresholds directly to the public proxy. These values are in proxy-premium units and are not the official one-basis-point WAPI/funding-rate threshold. At an eight-hour window, the three-basis-point activation rate is close to ten per cent.

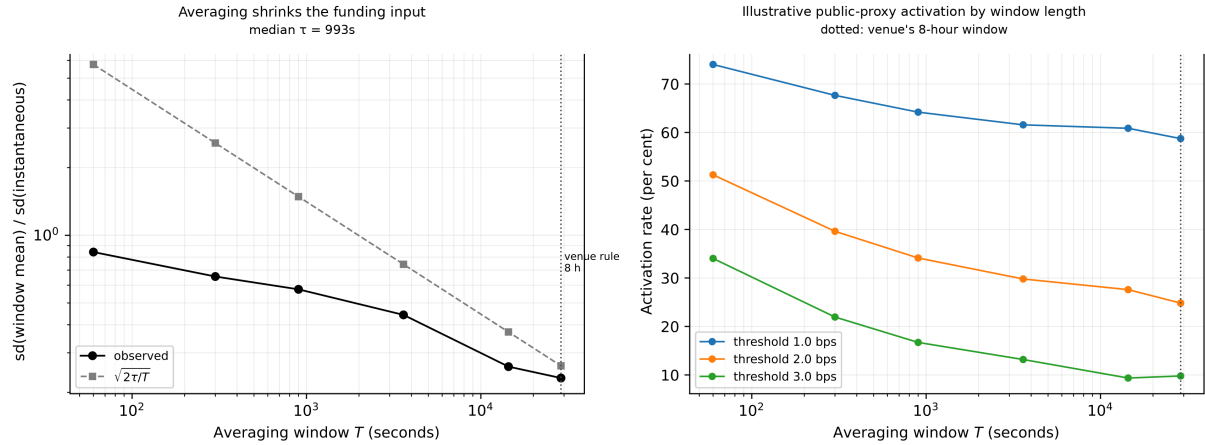


Figure 8: Averaging-horizon diagnostics. The dotted line marks eight hours. The one-, two-, and three-basis-point curves are illustrative thresholds applied directly to the public proxy; they are not official rule parameters. Calculations do not reconstruct the filed impact-price rule.

E Activation Robustness

The preferred activation curve uses 260 contract-intervals satisfying the complete-coverage rule. Figure 9 repeats the analysis for all 364 intervals with any usable coverage. The qualitative central inactive region remains, but partial intervals can bias the interval mean.

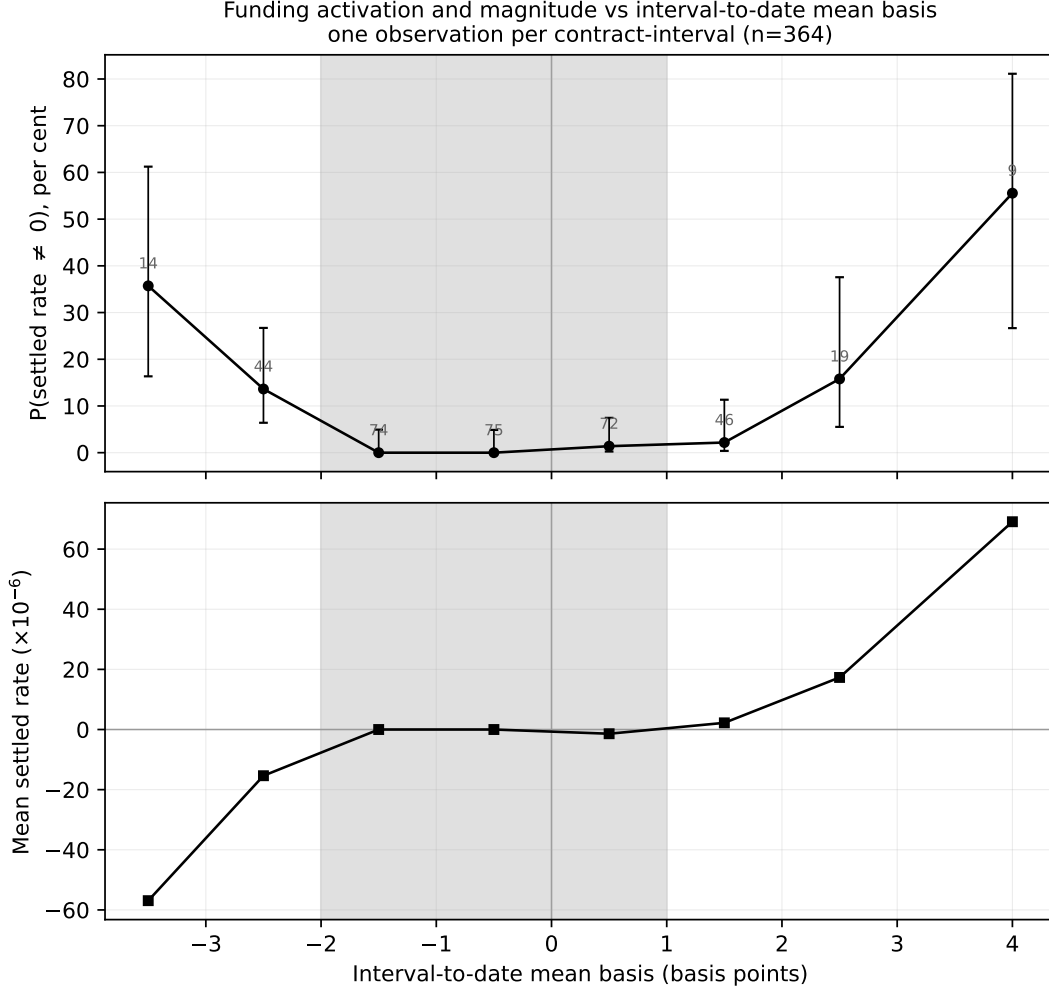


Figure 9: Funding activation and magnitude in the less restrictive sample of 364 contract-intervals. Any shaded central region is a descriptive public-proxy band, not the official WAPI/funding-rate threshold.

F Block-Bootstrap Results

Table 6 reports three-day circular block-bootstrap intervals for the exact-zero share. Each block retains all contracts observed at the resampled settlement events.

Table 6: Kalshi exact-zero shares in the matched window

Contract	n	Zero share (%)	Bootstrap 95% interval (%)
KXBCHPERP	95	62.1	35.8–85.3
KXBTCPERP	113	64.6	45.1–82.3
KXDOGEPERP	95	90.5	82.1–97.9
KXETHPERP	113	77.0	62.0–89.4
KXHYPEPERP	98	64.3	50.0–78.6
KXKSHIBPERP	95	54.7	33.7–76.8
KXLINKPERP	97	99.0	96.9–100.0
KXLTCPERP	95	67.4	49.5–84.2
KXNEARPERP	52	80.8	69.2–92.3
KXSOLPERP	113	92.9	87.6–97.4
KXSUIPERP	95	65.3	47.4–82.1
KXXRPPERP	113	56.6	35.4–77.0
KXZECPERP	52	80.8	61.5–96.2
Pooled	1,226	73.0	65.7–80.4

G Illustrative Offshore Thresholding

For BTCUSDT and ETHUSDT, the documented linearly weighted 480-minute Bybit Average Premium Index is reconstructed from minute observations (Bybit, 2026b). Applying an illustrative one-basis-point band directly to that weighted premium creates no zeros in 113 settlement-aligned observations for either asset. The result also holds for 899 rolling eight-hour windows per asset; the minimum absolute weighted premium is 2.782 basis points for BTC and 3.770 basis points for ETH.

This is an input-location diagnostic, not a structural Kalshi counterfactual. It omits Kalshi’s impact notional, sampling procedure, rate scaling, and treatment of unavailable depth.

H Additional Premium Diagnostics

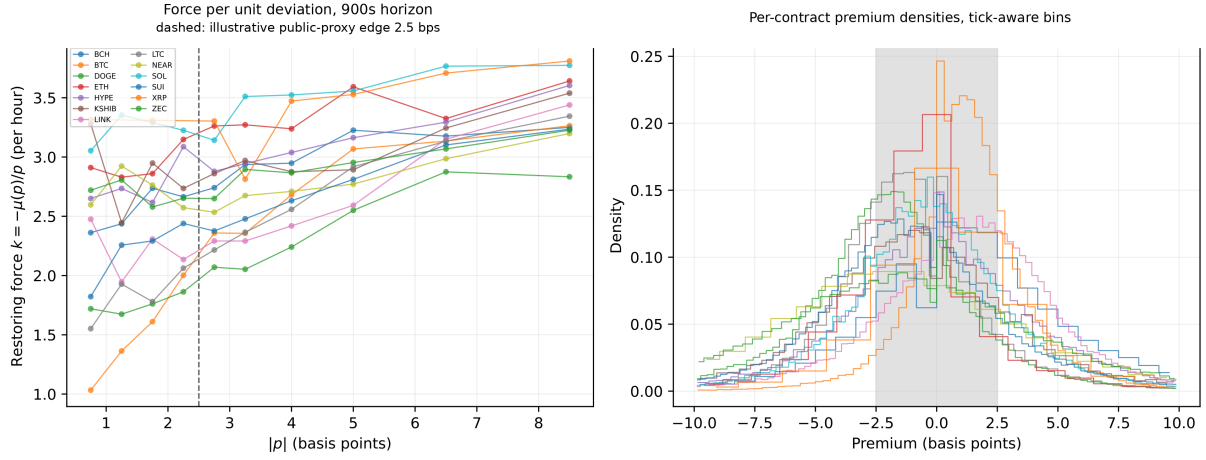


Figure 10: Restoring force per unit deviation and tick-aware contract-level premium densities. The 2.5-basis-point edge is an illustrative public-proxy value rather than the official one-basis-point WAPI/funding-rate threshold or a reconstruction of the filed rule.

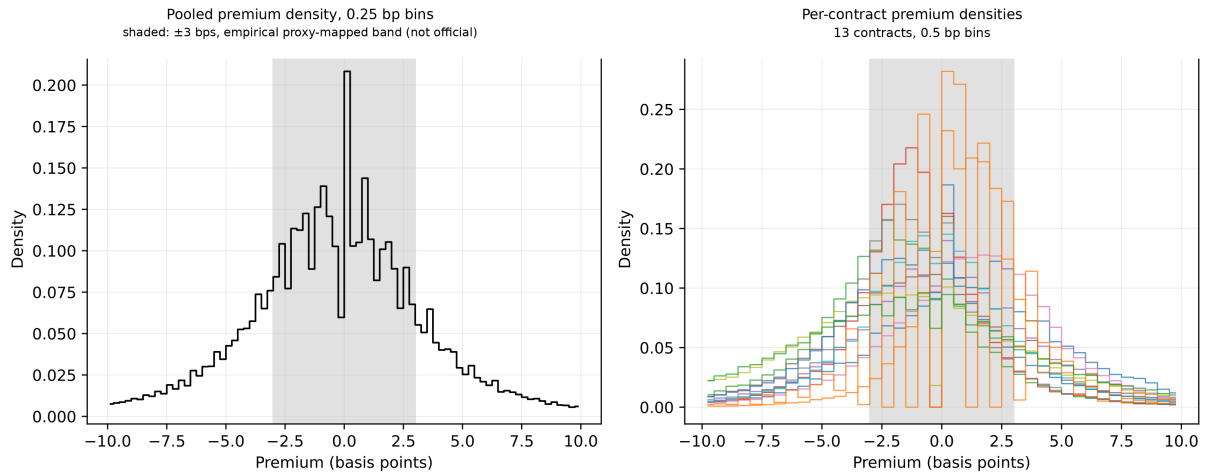


Figure 11: Pooled and contract-level public-premium distributions. The shaded region marks ± 3 basis points in public-proxy units, approximately the band obtained by mapping the official one-basis-point calculated-rate threshold through the empirical proxy slope; it is not the official deadband itself.

I Pseudo-Replication in the Funding-Estimate Stream

The public funding estimate is polled at sixty-second cadence. One settlement interval therefore contributes several hundred near-identical observations. Treating those observations as independent produces confidence intervals that are far too narrow and can create spurious local non-monotonicity. The analysis collapses the stream to one observation per contract-interval.

J Preregistered Holdout Design

A second collector began on 27 July 2026 on a supervised host. Before holdout outcomes were examined, a plan fixed the following analyses:

1. reconstruct the filed per-second impact-price Premium Index and chronologically weighted average as closely as sampling permits;
2. reconstruct the Help Center’s 480 one-minute candle description;
3. reconstruct the public mark-index proxy;
4. compare interval-level zero/nonzero confusion matrices, sign accuracy, and mean absolute rate error;
5. test threshold and cap fidelity with inference clustered by settlement timestamp;
6. conduct leave-one-contract-out sensitivity analysis;
7. apply refresh-time and lead-lag tests to endpoint alignment;
8. test price convergence around funding activation with directional predictions;
9. estimate realised-volatility splits and a parsimonious interaction with candidate-input magnitude; and
10. report funding-carry concentration and zero-run statistics without claiming execution-cost-adjusted arbitrage profitability.

The plan classifies the July sample as exploratory and prohibits confirmatory claims based on specifications selected after observing it.

K Data Files and Replication Materials

The replication materials include the matched Kalshi–Bybit CSV, summary tables, and bootstrap outputs used for the offshore comparison. The high-frequency relational store and compressed order-book archive are substantially larger and are available from the author upon reasonable request, subject to repository file-size limits and venue terms.